

(a) CPT data

Abstract: In any group of six people, there are always three mutual friends or three mutual strangers. We give a short proof ...

At a party of six, must some three guests all know each other, or some three be complete strangers? With five guests the answer is no, so six is the first case ...

We picture the six people as dots, joining friends with red lines and strangers with blue. The claim becomes one about colors ...

Pick any person: of the other five, some three are joined to her by the same color. Those three either all know one another or none of them do ...



unfold

writing request

Can you help me write a paper proving that among any six people there are three mutual friends or three mutual strangers?

plan and outlines

The proof is short, so the paper must first make the claim feel surprising. \$1 opens with the puzzle, \$2 draws it ...

pre-writing deliberation

Open with the party question so the reader feels the answer is not obvious ...

At a party of six, must some three guests all know each other, or some three be ...

pre-writing deliberation

Fix the dots-and-lines picture early, so every later step reads naturally ...

We picture the six people as dots, joining friends with red lines and strangers with blue. The claim becomes one about colors ...

... \$3, \$4 ...

pre-writing deliberation

The paper is done; now compress it. State the six-people fact, then the proof idea ...

In any group of six people, there are always three mutual friends or three mutual strangers. We give a short proof ...

(b) SFT data

We picture the six people as dots, joining friends with red lines and strangers with blue. The claim becomes one about colors ...

Pick any person: of the other five, some three are joined to her by the same color. Those three either all know one another or none of them do ...



reframe

task prompt

For a student talk: a known result says any six people include three mutual friends or three mutual strangers. Explain its dots-and-lines proof idea in under 100 words.

pre-writing deliberation

The audience is new to the topic, so keep the picture concrete and skip notation ...

Draw the six people as dots. Join two dots in red if they know each other, in blue if they do not. Now it is about colors ...

(c) PAW-Bench

A traffic light that stays green too long wastes time on an empty road, while a short green makes drivers stop and start again at every corner ...

We model the junction as a queue served in turns and ask which green-red cycle length keeps the average wait over all drivers smallest ...



examine

benchmark question

Read the attached paper and summarise the evidence behind the chosen cycle length, using only numbers it reports ...

We model the junction as a queue served in turns and ask which green-red cycle length keeps the average wait over all drivers smallest ...

scoring rubric

Covers the queueing assumption and the cycle length it implies (weight 0.4). Grounds every claim in the paper (0.3) ...

checklist

States the average-wait criterion. Uses no number absent from the paper. Stays within the requested length ...